

BCSE321L	MALWARE ANALYSIS			L	T	P	C
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Pre-requisite	NIL	Syllabus version					
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Course Objectives							
1. To introduce the malware taxonomy and malware analysis tools.							
2. To identify and analyze malware samples using static, dynamic analysis, and reverse engineering techniques.							
3. To detect and analyze malicious documents and mobile malware.							
Course Outcome							
After completion of this course, the student shall be able to:							
1. Possess the skills to carry out static and dynamic malware analysis on various malware samples.							
2. Understand the executable formats, Windows internals, and APIs.							
3. Apply techniques and concepts to unpack, extract, and decrypt malware.							
4. Comprehend reverse-engineering of malware and anti-malware analysis techniques.							
5. Achieve proficiency with industry-standard malware analysis tools.							
Module:1	Fundamentals of Malware Analysis					5 hours	
Malware taxonomy - Malware analysis techniques – Packed and Obfuscated Malware - Portable Executable File Format: Headers and Sections, Malware Analysis in Virtual Machines - Malware Analysis Tools: ProcMon/ ProcExplore, BinText, FileAlyzer, OllyDbg, etc.							
Module:2	Static Analysis					4 hours	
File signature analysis and Identifying file dependencies -Database of file hashes. String analysis - Local and online malware sandboxing - Levels of Abstraction - x86 Architecture - x86/x86_64 Assembly - Static Analysis Tools: PeiD, Dependency Walker, Resource Hacker.							
	Dynamic Analysis					4 hours	
Module:3							
Source level vs. Assembly level Debuggers - Kernel vs. User-Mode Debugging – Exceptions - Modifying Execution with a Debugger - Modifying Program Execution in Practice - DLL analysis - Dynamic Analysis Tools: Virustotal, Malware Sandbox, Windows Sysinternals							
Module:4	Reverse Engineering					4 hours	
Reverse engineering malicious code - Identifying malware passwords - Bypassing authentication -Advanced malware analysis: Virus, Trojan and APK Analysis - Reverse Engineering Tools: IDA Pro and OLLYDBG							
Module:5	Malicious Document Analysis					3 hours	
PDF and Microsoft Office document structures – Identify PDF and office document vulnerabilities - Analysis of suspicious websites - Examining malicious documents: word, XL, PDF, and RTF files - Malware extraction and analysis tools.							
Module:6	Anti-Reverse-Engineering					3 hours	
Anti-Disassembly - Anti-Debugging - Anti-Forensic Malware - Packers and Unpacking – Shellcode Analysis - 64-Bit Malware							
Module:7	Mobile Malware Analysis					5 hours	
Mobile application penetration testing - Android and iOS Vulnerabilities - Exploit Prevention - Handheld Exploitation - Android Root Spreading and Distribution Android							